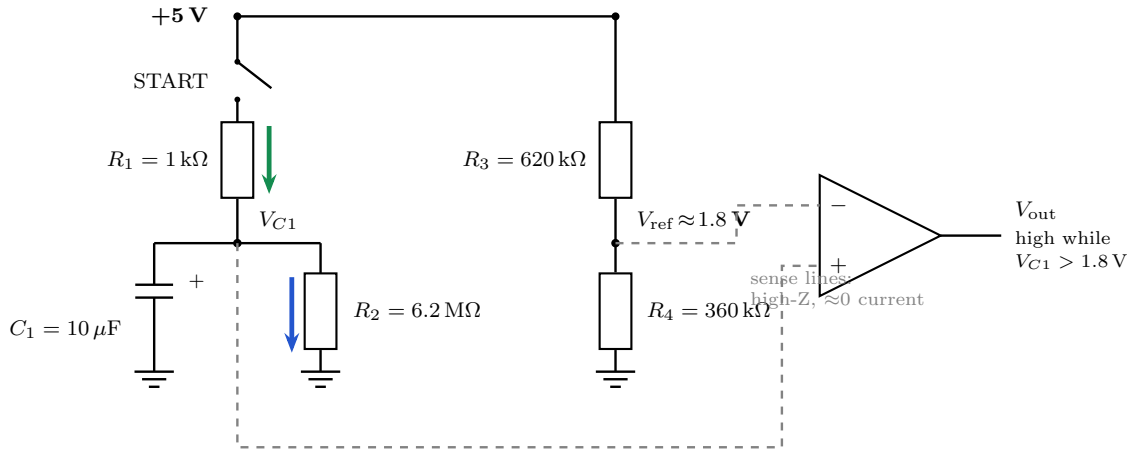




→ **charge** — down through R_1 (START pressed), $\tau \approx 10\ \text{ms}$

→ **discharge** — down through R_2 (START released), $\tau \approx 62\ \text{s}$

Why R_2 , not R_4 : R_4 sits on the V_{ref} node — a *separate* branch.

There is no conducting path from V_{C1} to R_4 , so C_1 can only empty through R_2 .